



PolarFire Xilinx Tradeoff

Issue 1.0

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PolarFire Xilinx Tradeoff

1. PolarFire and Xilinx accelerator trade-off

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Latency on both accelerators is included. Sample accuracy scores are not in this issue. The figures are qualitative.

Distribution:

- LMO (internal),
- European Space Agency (ESA)

Change Log:

- 0.1: First draft. Xilinx accuracy baseline from the CCN1 Design Justification File.
- 0.2: Deployment split, VectorBlox compile path, QAT weight selection, Xilinx and PolarFire latency, qualitative overlays.

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2. Introduction

The DIOSSA RPO payload estimates the 6-degree-of-freedom pose of a target spacecraft from monocular images. The vision chain is two convolutional networks in series: FCOS for object detection, then MobilePose for keypoint regression. In CCN1 those networks were trained in floating point, refined with quantization-aware training, and compiled onto the Xilinx Deep Learning Processing Unit (DPU) in an UltraScale+ MPSoC through Vitis-AI 3.5.

This note compares that baseline with the same networks compiled for the Microchip PolarFire SoC and its VectorBlox accelerator (SDK 3.1). The networks and the CCN1 weights stay the same. What changes is the accelerator and the compile path.

2.1. Objective

Show that the CCN1 networks can be compiled and executed on PolarFire, and put the measured latency next to the Xilinx DPU and CPU-head measurements from the earlier benchmarking campaign.

Scored accuracy on the PolarFire sample (IoU, keypoint error) is not part of this issue. The CCN1 validation numbers remain the accuracy record. A few overlays are included so the running chain can be seen.

2.2. Scope

- Object detection: FCOS, input 384×288 , checkpoint `quare-delf` epoch 19.
- Keypoint regression: MobilePose, input 224×224 , 20 keypoints, checkpoint `bijou-rasp` epoch 17.

A second FCOS resolution, 640×480 , was trained in CCN1. It is quoted only as context. It was not compiled for PolarFire.

2.3. Applicable Documents

AD-n	Document Title	Issue
AD-1	PL23-0005-DJF-0001 i1.1 (DIOSSA-CCN1)	1.1

2.4. Reference Documents

RD-n	Document Title
RD-1	VectorBlox SDK, Tutorial Walkthrough Guide (third-party/vbx-sdk/docs/tutorial_walkthrough_guide.md)
RD-2	Microchip VectorBlox Accelerator SDK, product description of the 3.1 release

2.5. Definitions

Acronym	Meaning
DPU	Xilinx Deep Learning Processing Unit
EDGE	The part of the network compiled onto the accelerator
FLOAT	The float32 network, evaluated on a GPU
FCOS	Fully Convolutional One-Stage object detector

Acronym	Meaning
QAT	Quantization-aware training. Weights are stored as float32, but they were trained under a fake-quantization constraint.
VNNX	VectorBlox binary executed by the accelerator

3. How the network is split

Following the previous DIOSSA phases, each network is split in two.

The first part runs on the FPGA accelerator. It is kept as large as the core will accept, so the convolutional trunk uses the accelerator and the CPU is left with as little as possible.

The second part stays on the CPU. It is the tail that cannot be placed on the accelerator. For VectorBlox that tail is whatever does not survive the path to a full-integer INT8 TensorFlow Lite graph and then to `vnnx_compile`. The reasons a layer is left behind are:

- the operator is outside the INT8 TensorFlow Lite set that VectorBlox compiles [RD-1];
- the layer is post-processing rather than a convolution. The SDK tutorials cut this off on purpose (`tf_lite_cut` on YOLOv8) before compilation: box decoding, non-maximum suppression, top-k, and the reduction from a heatmap to a coordinate;
- the tensor shape is dynamic, or an attribute of the operator (rank, dilation, alignment) is not accepted by the compiler;
- the activation memory of the whole graph does not fit the selected core. The build used here is the V1000 configuration with no compression.

On PolarFire the CPU tail is an ONNX graph executed by ONNX Runtime on the RISC-V application cores. Accelerator outputs are dequantized on the CPU and then passed into that graph. FCOS consumes the box, class, and centerness maps and emits boxes, scores, and labels. MobilePose consumes the heatmaps and emits coordinates. The same split was used on the Xilinx side: the DPU runs an `.xmodel`, and the post-processing head runs as ONNX on the MPSoC CPU.

4. Embedding a PyTorch model on PolarFire

VectorBlox does not execute PyTorch. The published SDK 3.1 path for an ONNX model is: convert to TensorFlow Lite, quantize to INT8 if the file is not already quantized, insert preprocessing with `tflite_preprocess`, and compile with `vnnx_compile` [RD-1], [RD-2]. The DIOSSA builds follow that path. The steps actually run are:

1. Load the PyTorch checkpoint and export two ONNX files. One is the accelerator trunk. The other is the CPU head.
2. Build a calibration sample. One hundred images are taken from the CCN1 test sample, resized to the network input, and stored as a NumPy array. The array is normalized with the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).
3. Simplify the ONNX graph with `onnxsim`, so the converter sees a static graph without the training-time debris.
4. Convert ONNX to full-integer INT8 TensorFlow Lite with `onnx2tf`. This step both translates the graph and quantizes it. The calibration array is what the quantizer uses to pick a scale and a zero-point per tensor.
5. Insert preprocessing with `tflite_preprocess`. Mean and scale are written into the graph, scaled into the 0–255 pixel range, so the accelerator consumes the image with the same normalization the network was trained with. The tool also inserts the uint8-to-int8 quantize at the input.
6. Compile with `vnnx_compile -s V1000 -c ncomp`. The output is a `.vnnx` binary for the V1000 core, without weight compression.

`vnnx_compile` is the step that maps the INT8 graph onto the VectorBlox core. Anything it rejects stays in the ONNX head from step 1.

5. Why the QAT checkpoints were used

The first attempt was to embed the CCN1 float checkpoints (`licit-weal` for FCOS, `blest-harl` for MobilePose). That quantization did not complete. `onnx2tf` could not assign a scale and a zero-point on those weights, so no INT8 TensorFlow Lite file was produced and there was nothing to compile.

The weights that did quantize are the CCN1 quantization-aware checkpoints, `quare-delf` and `bijou-rasp`. They are still stored as float32. They were trained with Vitis-AI fake quantization, so the values already sit on a grid that post-training quantization can represent. With those weights the same calibration sample produces a finite scale and zero-point, and the compile goes through.

This is the same pair that was embedded on the Xilinx DPU in CCN1. The float parents are the accuracy reference. They are not the binaries on either accelerator.

6. Xilinx accuracy baseline

The figures in this section are copied from AD-1. They were measured on the CCN1 validation set. They are not re-measured in this issue, and they are not PolarFire numbers. Inference rates quoted from AD-1 were measured on a laptop RTX 3070, not on the DPU.

6.1. Object detection (FCOS)

FCOS was selected over Faster R-CNN in CCN1. On a 480×640 input the float FCOS reached IoU 0.874 and mAP at IoU 0.75 of 0.956, against 0.853 and 0.901 for Faster R-CNN, at 45.78 FPS against 37.66 FPS on the RTX 3070, with 32.1 M parameters against 43 M. Faster R-CNN cannot be trained with the Vitis-AI quantization-aware procedure, so it was dropped.

The embedded resolution is 384×288 . The best float model at that size is `licit-weal` (IoU 0.925). Quantization-aware training from that checkpoint produced `quare-delf` (IoU 0.858). The loss accepted in CCN1 is 0.067 IoU. The 640×480 QAT run lost more (IoU 0.958 down to 0.819) and was not the network embedded for the shorter approach.

Table 6.1 FCOS, 384×288 . Validation metrics from AD-1.

	FLOAT <code>licit-weal</code>	QAT <code>quare-delf</code>
Started from	COCO	licit-weal
Input [W × H]	384×288	384×288
IoU	0.925	0.858

	FLOAT licit-weal	QAT quare-delf
mAP	0.858	0.693
mAP at IoU > 0.50	0.989	0.978
mAP at IoU > 0.75	0.946	0.850
RTX 3070 throughput	65.12 FPS	not reported

6.2. Keypoint regression (MobilePose)

The float model selected in CCN1 is **blest-harl**, initialised from the DIOSSA-1 MobilePose checkpoint and trained on 20 keypoints. Raising the input from 224×224 to 448×448 did not improve accuracy. Predicting all 41 keypoints was worse than the 20-keypoint subset. **blest-harl** is reported at 3.799 L2 pixel error. AD-1 does not repeat whether that L2 is on the 2590×1942 camera image or on the network input. The QAT section does give the float L1 on the camera image: 4.783 px.

The QAT model is **bijou-rasp**, started from **blest-harl**. L1 on the camera image is 5.303 px. The accepted quantization loss is 0.520 px. Two other learning rates were worse (**smoky-quey** 6.848 px, **swish-mows** 5.797 px).

Table 6.2 MobilePose, 20 keypoints. Metrics from AD-1.

	FLOAT blest-harl	QAT bijou-rasp
Started from	DIOSSA-1 MobilePose	blest-harl
Input	224×224	224×224
Keypoints	20	20
L2 pixel error (reference frame not stated)	3.799	—

	FLOAT blest-har1	QAT bijou-rasp
L1 on the camera image [2590 × 1942], px	4.783	5.303
L1 on the network input, px	not stated	1.564
L2 on the camera image [2590 × 1942], px	not stated	4.216
L2 on the network input, px	not stated	1.238

On the DPU, AD-1 reports the end-to-end pose rather than a separate IoU or keypoint error for the compiled binary. Relative position error of the QAT pair on the DPU stays under 2% at 1-sigma on trajectories 2 and 3. A rotation error near 180 degrees on trajectory 1 is attributed to symmetry of the keypoints and is already present on the float models.

7. Latency

Both platforms run the same split: the convolutional trunk on the accelerator, the head on the CPU through ONNX Runtime. The times below are those two pieces, measured separately and then added. They are not one timed capture of the whole payload loop. Image decode and the pose solver are not included.

7.1. Xilinx

The Xilinx numbers come from the earlier campaign on the MPSoC ([reports/xilinx-latency-benchmarks](#)). The DPU time is [xdputil benchmark](#) with one thread, run for 60 seconds. The CPU time is [onnxruntime_perf_test](#) in duration mode for 20 seconds, on the ONNX head. The thread count of that ONNX run was not pinned.

The rows used here are the embedded shapes. MobilePose is [mobilepose_224x224_20k](#) (224 × 224, 20 keypoints). FCOS is [fcos_288x384_2c](#) (height 288, width 384, two classes). Other shapes in those files are not the binaries under comparison.

Table 7.1 Xilinx MPSoC. DPU time and ONNX Runtime head, measured separately.

	DPU	CPU head	Sum
MobilePose 224 × 224	2.95 ms (338.5 FPS)	56.84 ms	59.8 ms (16.7 FPS)
FCOS 384 × 288	51.81 ms (19.30 FPS)	4.48 ms	56.3 ms (17.8 FPS)

On the DPU, MobilePose is a small fraction of the chain. Almost all of the summed time is the ONNX head. FCOS is the opposite: the DPU is the slow piece, and the head is under 5 ms.

7.2. PolarFire

PolarFire times were measured on the board at 192.168.20.6, VectorBlox V1000, no compression. Each network was run on 50 images from the CCN1 sample, one loop per image. The standard deviation of the accelerator time is 0.01 ms for MobilePose and 0.02 ms for FCOS, so the mean is the whole story. CPU post-processing is dequantization plus ONNX Runtime. ONNX Runtime was run with one intra-op thread.

Table 7.2 PolarFire VectorBlox. Mean over 50 images.

Metric	MobilePose 224 × 224	FCOS 384 × 288
Accelerator	17.20 ms (58.2 FPS)	394.67 ms (2.53 FPS)
Dequantize	4.33 ms	1.81 ms
ONNX Runtime	16.80 ms	13.37 ms
CPU post-processing	21.13 ms	15.18 ms
End to end	38.33 ms (26.1 FPS)	409.85 ms (2.44 FPS)

7.3. Comparison

Table 7.3 Accelerator time and end-to-end time.

	Xilinx DPU	PolarFire VectorBlox	Ratio, PolarFire / Xilinx
MobilePose accelerator	2.95 ms	17.20 ms	5.8
MobilePose end to end	59.8 ms (sum)	38.33 ms	0.64
FCOS accelerator	51.81 ms	394.67 ms	7.6
FCOS end to end	56.3 ms (sum)	409.85 ms	7.3

MobilePose on the VectorBlox core is slower than on the DPU (17.2 ms against 2.95 ms). The ONNX head on the PolarFire RISC-V cores is faster than the head measured on the MPSoC (21.1 ms against 56.8 ms). Added together, the PolarFire chain is the shorter one: 38.3 ms against 59.8 ms.

FCOS does not follow that pattern. The VectorBlox core takes 395 ms, about 7.6 times the DPU, and the head does not compensate. End to end is 410 ms, 2.4 frames per second, against 56 ms on the Xilinx sum. Closing that gap is the open performance item. The two measurements were not made with the same harness, so the ratio is the right precision to claim, not a tenth of a millisecond.

8. Qualitative overlays

The figures below are PolarFire outputs on the CCN1 Inmarsat-5 sample, drawn on the network input (224 × 224 for MobilePose, 384 × 288 for FCOS). Ground truth is green. The prediction is red. These frames are examples. They are not a score.

8.1. Keypoint regression

On these three frames the red keypoints sit on the same structure as the green ones: body, antennae, and nozzle. That is the behaviour the CCN1 QAT model had on the DPU. It is not a measured L1.

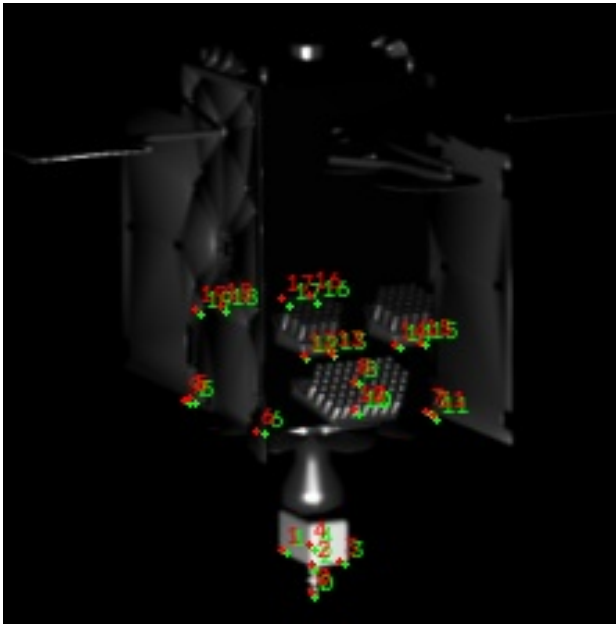


Fig. 8.1 MobilePose, `VisCam_0021021` . Green: ground truth. Red: PolarFire.

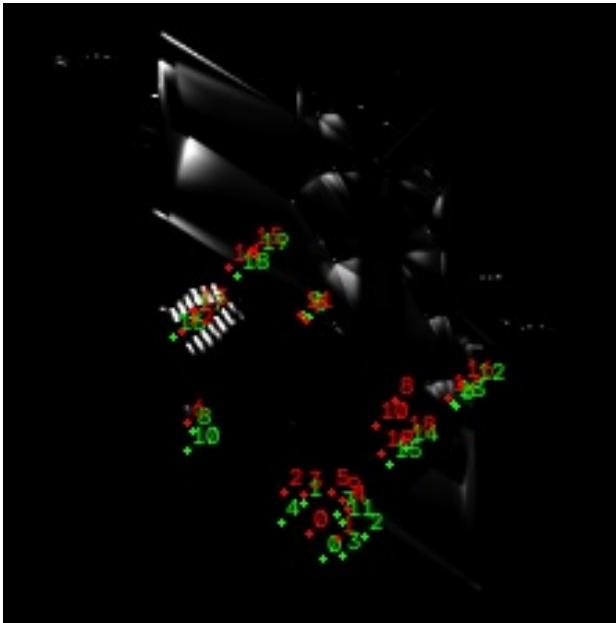


Fig. 8.2 MobilePose, `VisCam_0001775` . Green: ground truth. Red: PolarFire.

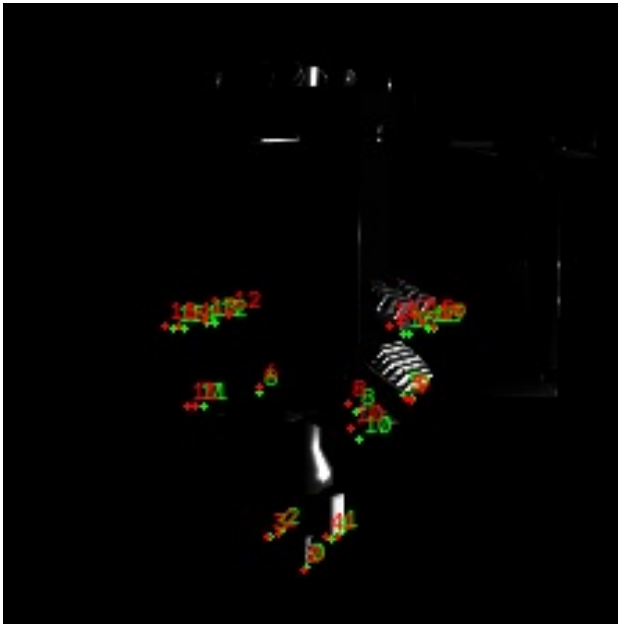


Fig. 8.3 MobilePose, `VisCam_0016747`. Green: ground truth. Red: PolarFire.

8.2. Object detection

On these frames the network returns a detection, and the red box falls inside the spacecraft. The box is much smaller than the green ground-truth extent. That is visible here and it is not scored in this issue. It has to be explained before detection can be called equivalent to the CCN1 QAT result (IoU 0.858 on the validation set).

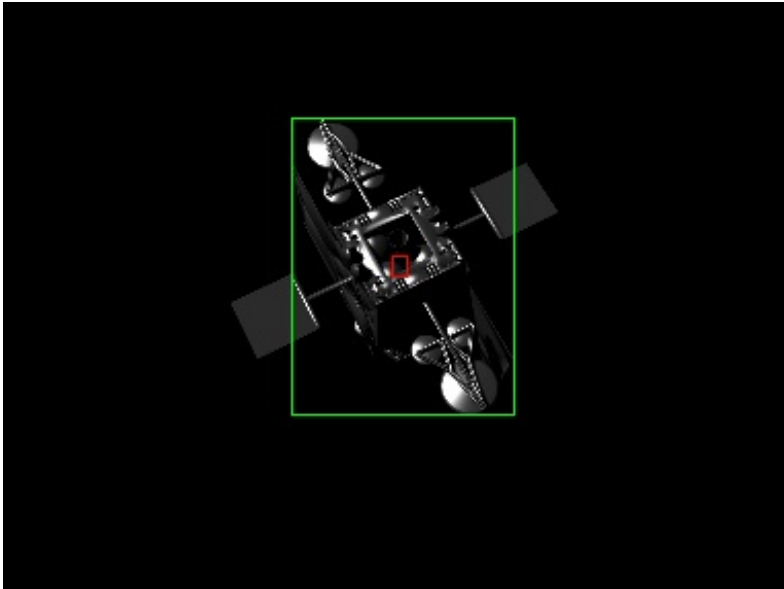


Fig. 8.4 FCOS, `VisCam_0002932`. Green: ground truth. Red: PolarFire.

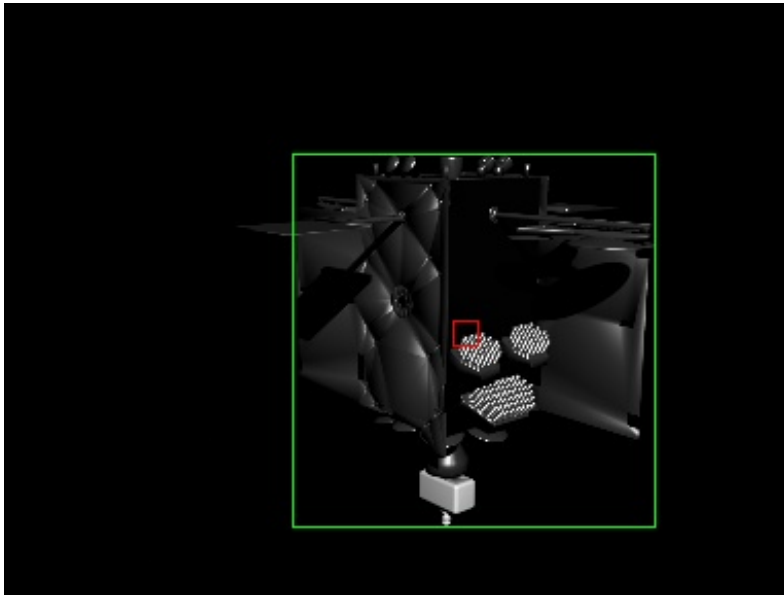


Fig. 8.5 FCOS, `VisCam_0008192`. Green: ground truth. Red: PolarFire.

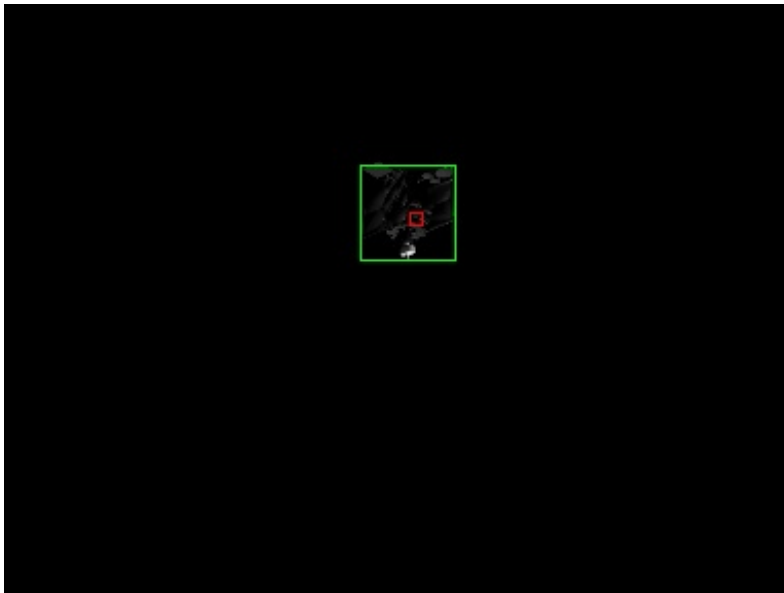


Fig. 8.6 FCOS, `VisCam_0031435`. Green: ground truth. Red: PolarFire.

9. Conclusion

The CCN1 QAT checkpoints compile for VectorBlox and run on the PolarFire SoC, with the convolutional trunk on the accelerator and the head on ONNX Runtime. The float checkpoints do not quantize. The QAT weights do, which is why the switch uses `quare-delf` and `bijou-rasp` rather than `licit-weal` and `blest-harl`.

On latency, MobilePose end to end is shorter on PolarFire than the summed Xilinx measurement (38 ms against 60 ms). FCOS is not: the VectorBlox core is about eight times the DPU time, and the chain runs at 2.4 frames per second.

On the overlays, MobilePose keypoints follow the ground truth on the frames shown. FCOS places a box on the spacecraft, and that box does not cover the ground-truth extent. Neither observation replaces a scored comparison against the Xilinx edge binaries on this sample.

10. Open items

1. Score FCOS and MobilePose on the shared CCN1 sample, PolarFire against the Xilinx edge binaries, and report IoU and keypoint error.
2. Account for the FCOS accelerator time (395 ms on VectorBlox against 52 ms on the DPU): operator coverage, the V1000 configuration, and how much of the trunk actually landed on the core.
3. Account for the detection boxes in the figures, which sit inside the ground-truth box and do not match its extent.

